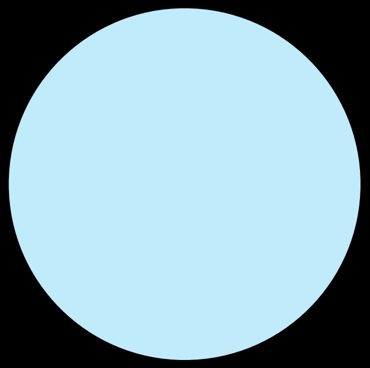
AA/EE/ME 548

Linear Multivariable Control

Signal temporal logic & human robot interactions

Spring 2026

Spatial (state) constraints
E.g., Avoid obstacle, velocity limits



Spatial (state) constraints
E.g., Avoid obstacle, velocity limits

What about temporal constraints?

Between time $[a, b]$ be in region A and eventually reach region B

Go to region A before going to region B

If C happens, then D must happen and always A must hold

How to write these constraints (specifications) mathematically?

Signal temporal logic

Formal logic: A "mathematical language" (words + grammar = sentence with meaning) that can describe spatio-temporal properties

$\phi ::=$	$\top$	μ_c	$\neg\phi$	$\phi \wedge \psi$	$\phi\,\mathcal{U}_{[a,b]}\,\psi$
	True	Predicate	Not	And	Until

"Sentences" are defined recursively using the grammar

A logic language for specifying spatio-temporal properties of *signals* (continuous time, real-valued numbers).

$$\mathbf{x} = x_0, x_1, \ldots, x_T \quad \text{(discretize time)}$$

https://arxiv.org/abs/2501.04194

Signal temporal logic: Boolean semantics

Boolean Semantics

$$
\begin{aligned}
\mathbf{x} \models \mu_c \quad &\Leftrightarrow \quad \mu(x_0) > c \\
\mathbf{x} \models \neg\phi \quad &\Leftrightarrow \quad \neg(\mathbf{x} \models \phi) \\
\mathbf{x} \models \phi \wedge \psi \quad &\Leftrightarrow \quad (\mathbf{x} \models \phi) \wedge (\mathbf{x} \models \psi) \\
\mathbf{x} \models \Diamond_{[a,b]}\phi \quad &\Leftrightarrow \quad \exists t \in [a,b] \ \text{ s.t. } \mathbf{x}_t \models \phi \\
\mathbf{x} \models \Box_{[a,b]}\phi \quad &\Leftrightarrow \quad \forall t \in [a,b] \ \text{ s.t. } \mathbf{x}_t \models \phi \\
\mathbf{x} \models \phi\, \mathcal{U}_{[a,b]}\, \psi \quad &\Leftrightarrow \quad \exists t \in [a,b] \ \ s.t. \ (\mathbf{x}_t \models \psi) \\
&\qquad \wedge \ (\forall \tau \in [0,t], \mathbf{x}_\tau \models \phi)
\end{aligned}
$$

Signal temporal logic: Robustness

STL is also equipped with *quantitative semantics*, a measure of *how much* the signal satisfies/violates the specification

Quantitative Semantics (Robustness Formulas)

$$
\begin{aligned}
\rho(\mathbf{x}, \top) &= \rho_{\max} \qquad \text{where } \rho_{max} > 0 \\
\rho(\mathbf{x}, \mu_c) &= \mu(x_0) - c \\
\rho(\mathbf{x}, \neg\phi) &= -\rho(\mathbf{x}, \phi) \\
\rho(\mathbf{x}, \phi \wedge \psi) &= \min(\rho(\mathbf{x}, \phi), \rho(\mathbf{x}, \psi)) \\
\rho(\mathbf{x}, \Diamond_{[a,b]}\phi) &= \max_{t \in [a,b]} \rho(\mathbf{x}_t, \phi) \\
\rho(\mathbf{x}, \Box_{[a,b]}\phi) &= \min_{t \in [a,b]} \rho(\mathbf{x}_t, \phi)
\end{aligned}
$$

$$
\rho(\mathbf{x}, \phi \mathcal{U}_{[a,b]} \psi) = \max_{t \in [a,b]} \left\{ \min \left(\min_{\tau \in [0,t]} \rho(\mathbf{x}_\tau, \phi), \rho(\mathbf{x}_t, \psi) \right) \right\}. \tag{3}
$$

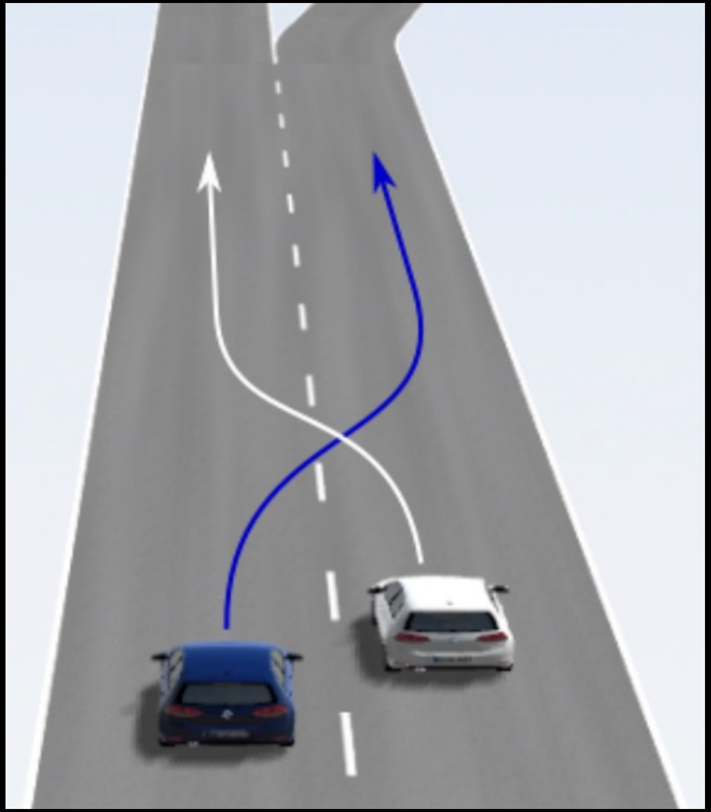
Signal Temporal Logic:

A way to convert spatio-temporal specifications into a scalar value

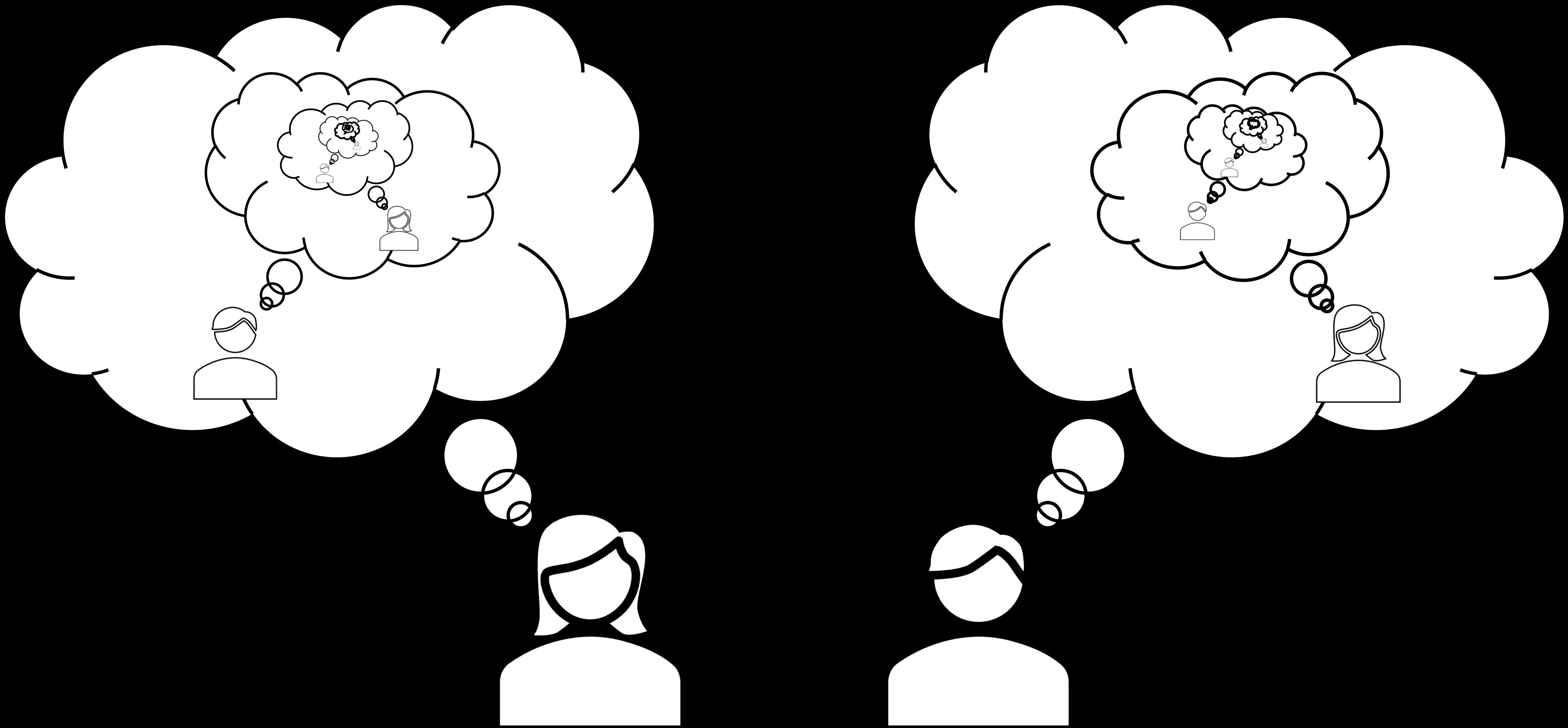
$$\phi = \square_{[aT,bT]}(\text{inside } {\color{orange}\bullet}) \wedge \Diamond(\text{inside } {\color{green}\bullet})$$

Planning around humans

Multimodal uncertainty



Schmerling et al 2018



Need to account for how others may respond to your own actions

What can we do?

Some references

- https://msl.stanford.edu/papers/wang_game-theoretic_2021.pdf
- https://www.youtube.com/watch?v=589YHBMSFjQ
- https://arxiv.org/abs/2404.03734
- https://arxiv.org/abs/1710.09483
- https://arxiv.org/abs/2012.03390